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## H1 Biology

Paper 1 Multiple Choice

**8876/01**

**20 September 2018**

**60 minutes**

Additional Materials: Multiple Choice Answer Sheet

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### READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use paper clips, glue or correction fluid.

Write your name, civics group and registration number on the Answer Sheet in the spaces provided.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

**Read the instructions on the Answer Sheet very carefully.**

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

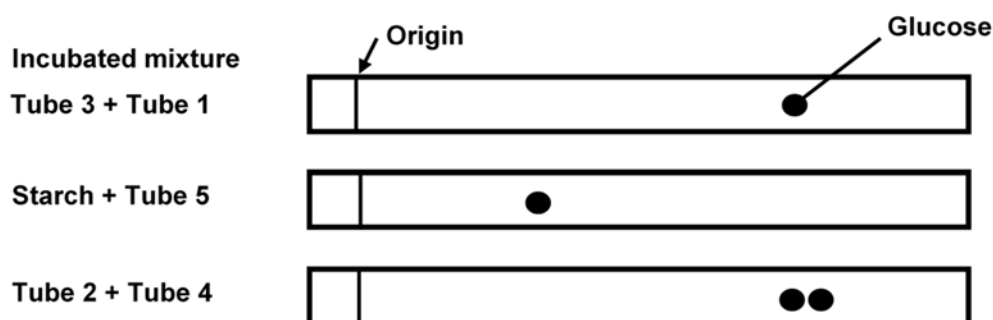
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This document consists of **23** printed pages and **1** blank pages.

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- 1 A lab technician has 5 tubes containing unknown compounds. However, he knows that each tube contains only either a carbohydrate or enzyme. To determine the identities of the compounds in the 5 tubes, the lab technician conducted various tests and he made the following observations.

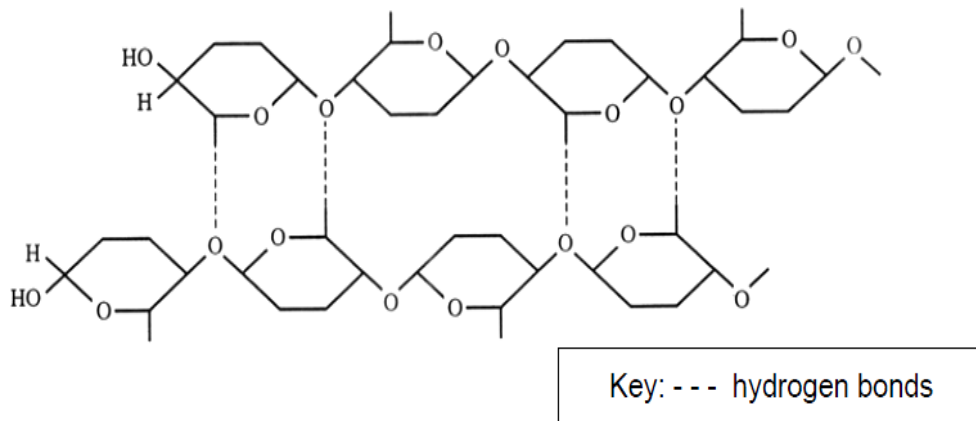
- I Sample in tube 2 is soluble in water. Sample in tube 3 was insoluble in water.
- II Tube 5 tested positive with Biuret's test.
- III Contents of all 5 samples tested negative with Benedict's test.
- IV However, after he mixed and incubated the contents of certain tubes together, a positive result was obtained for Benedict's test. The mixtures were also subjected to paper chromatography, and the results were shown below.



Which of the following correctly shows the contents of each tube?

|   | tube 1    | tube 2    | tube 3    | tube 4    | tube 5    |
|---|-----------|-----------|-----------|-----------|-----------|
| A | amylase   | cellulose | sucrose   | cellulase | sucrose   |
| B | cellulase | sucrose   | cellulose | sucrase   | amylase   |
| C | cellulose | sucrose   | cellulose | sucrase   | amylase   |
| D | sucrose   | cellulose | sucrase   | amylase   | cellulase |

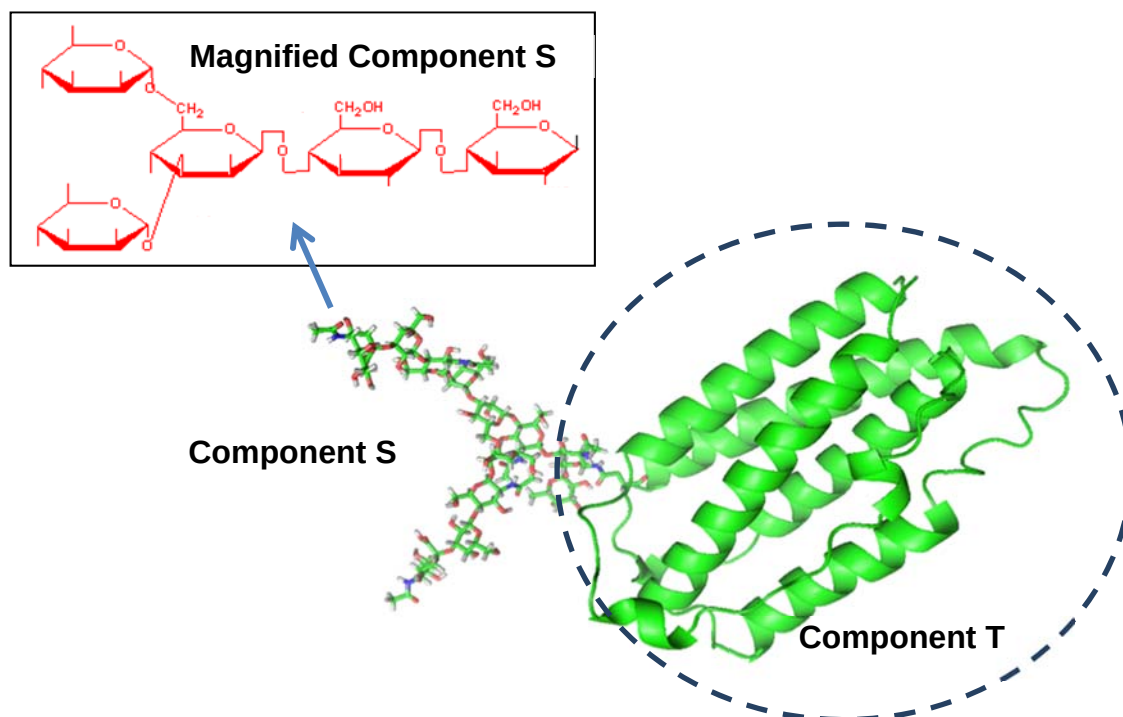
- 2** The figure below shows part of the molecular structure of a polysaccharide.



Which of the following statements are true?

- I** The hydrogen bonds maintain the double-helical structure of the molecule.
  - II** The subunits of each chain are  $\beta$ -glucose.
  - III** Water is needed in the formation of bonds between the subunits.
  - IV** The chains formed by the subunits are unbranched.
- A** I and II  
**B** II and IV  
**C** III and IV  
**D** All of the above

- 3** The figure below shows the structure of a biomolecule extracted from a cell. It is made up of two components, **S** and **T**. The boxed region shows a **magnified** portion of component **S**. The circled region shows component **T**, of the same molecule.

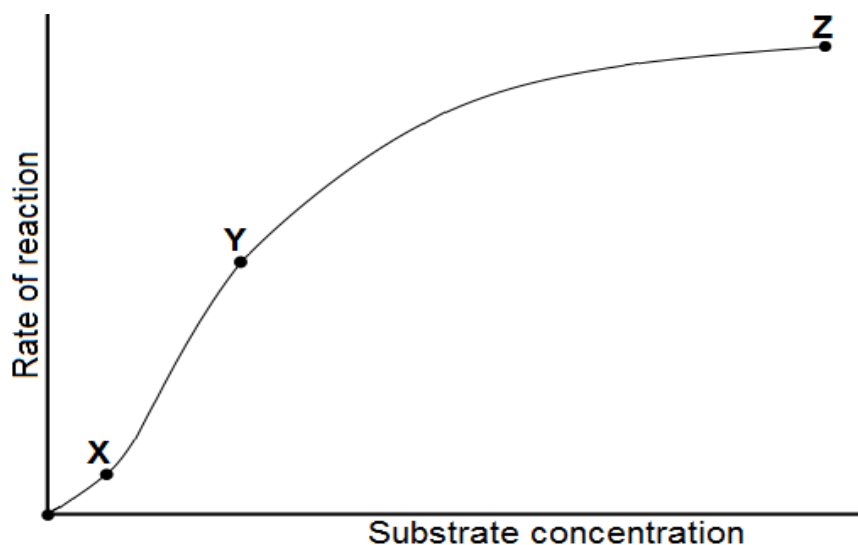


Below are some statements regarding the structure, property and function of the biomolecule extracted from the cell as shown in the figure above. Which of these statements are true?

- I The biomolecule has both hydrophilic and hydrophobic properties.
- II It is important for cell to cell adhesion.
- III The synthesis of this molecule requires only the rough endoplasmic reticulum.
- IV When completely hydrolysed, all the monomers of this molecule are soluble in water.

- A I and III
- B II and IV
- C I, II and IV
- D II, III and IV

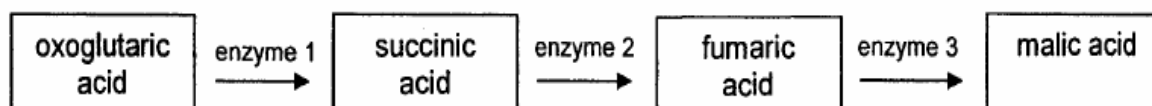
- 4 The graph below shows the effect of increasing substrate concentrations on the activity of an allosteric enzyme under optimum conditions.



Which of the following statements is correct?

- A There is low kinetic energy at X to overcome the activation energy, thus resulting in a low rate of reaction.
- B Rate of reaction increases at a faster rate at Y as the allosteric activator outcompetes the allosteric inhibitor to bind to the allosteric site.
- C At Z, enzyme molecules are in the active state and active sites are saturated.
- D Substrate concentration is the limiting factor at X and Y but temperature is the limiting factor at Z.

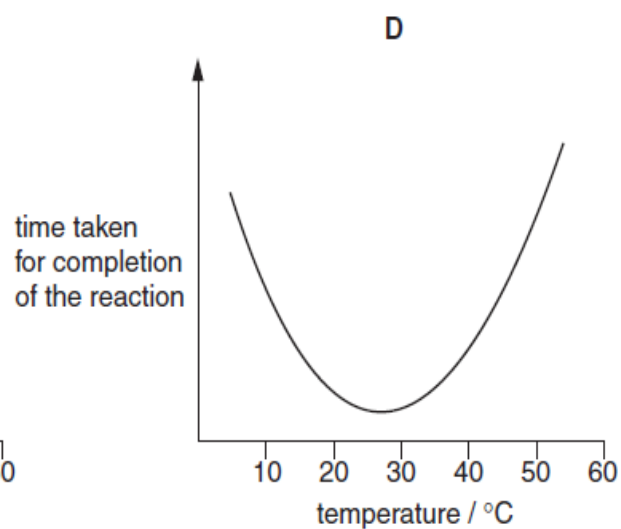
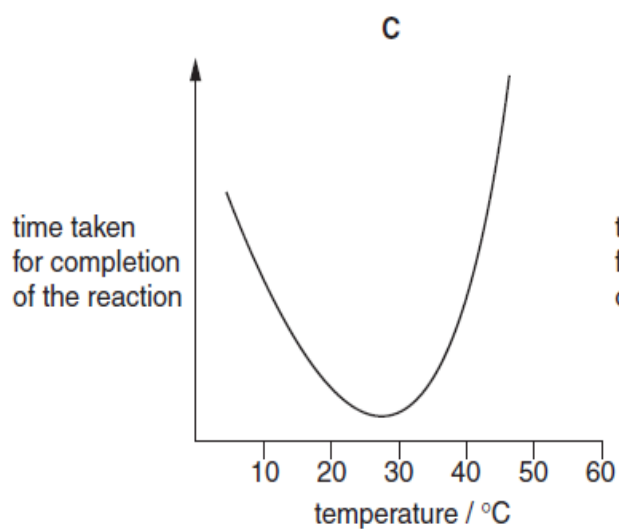
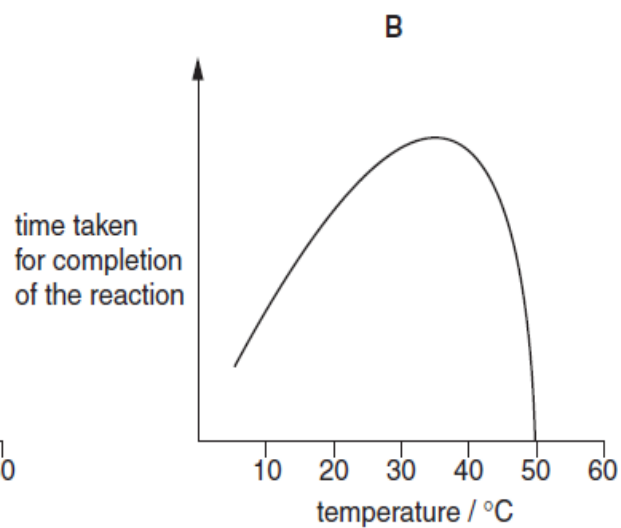
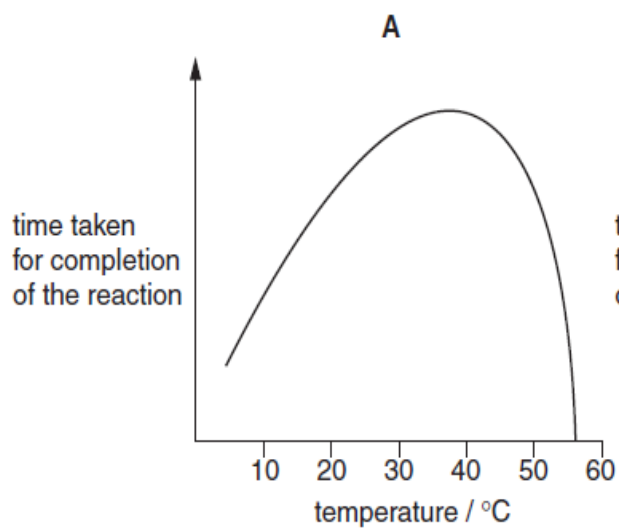
- 5 The four acids shown below form part of an enzyme-catalysed sequence of reactions. The addition of malonic acid results in no change in the concentration of oxoglutaric acid, an accumulation of succinic acid, and a near absence of both fumaric acid and malic acid. Further addition of fumaric acid results in the formation of malic acid. What does this information indicate about malonic acid?



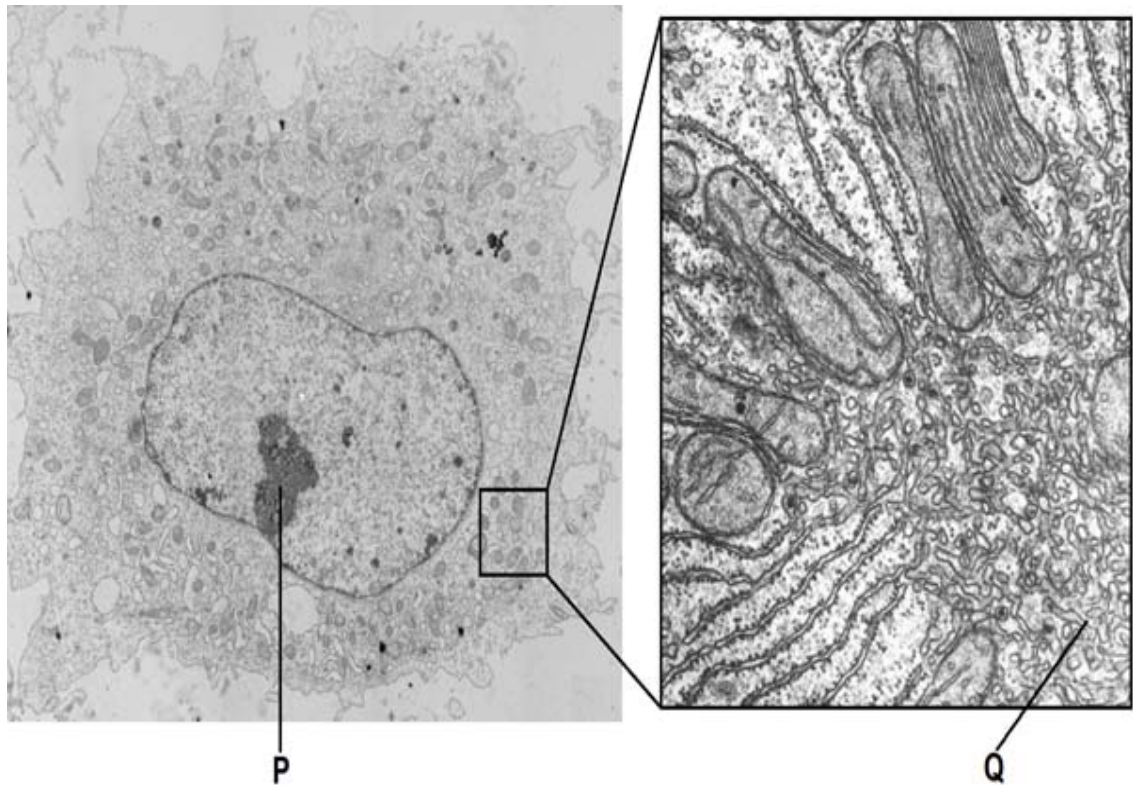
- A It is an inhibitor of enzyme 1.
- B It is an inhibitor of enzyme 2.
- C It reactions with fumaric acid.
- D It catalyses the formation of succinic acid.

- 6 An enzyme is completely denatured at 50°C. A fixed concentration of this enzyme is added to a fixed concentration of its substrate. The time taken for completion of the reaction is measured at different temperatures.

Which graph shows the results?



- 7 The diagram on the left below shows the electron-micrograph of a cell found in a healthy individual. A region of the cell, indicated by the box, is magnified and shown in the diagram on the right.



Which of the following shows the correct identities and functions of **P** and **Q**?

|          | Identity of P | Function of P                  | Identity of Q      | Function of Q               |
|----------|---------------|--------------------------------|--------------------|-----------------------------|
| <b>A</b> | Nucleosome    | Transcription of rRNA          | Smooth ER          | Detoxification              |
| <b>B</b> | Nucleosome    | Assembly of ribosomal subunits | Secretory vesicles | Storage of $\text{Ca}^{2+}$ |
| <b>C</b> | Nucleolus     | Assembly of ribosomal subunits | Smooth ER          | Synthesis of lipids         |
| <b>D</b> | Nucleolus     | Transcription of rRNA          | Secretory vesicles | Modification of protein     |



- 8** Most wild plants contain toxins that deter animals from eating them. A scientist discovered that a toxin produced by a certain plant was also toxic to the same plant if it is applied to the roots of the plant.

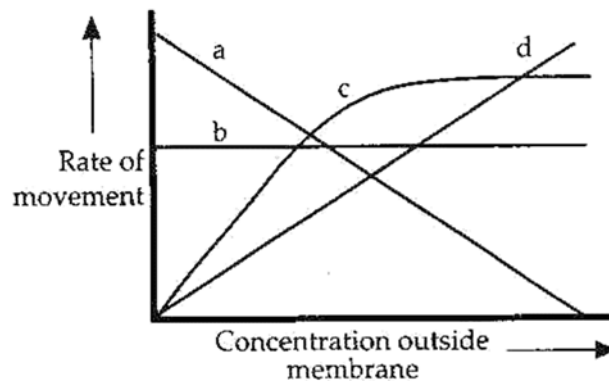
As the first step on finding out why the plant was not normally killed by its own toxin, he fractionated some plant cells and found that the toxin was in the fraction that contained the largest cell organelle. He also found that the toxin was no longer toxic after it was heated.

Which of the following statements are consistent with the scientist's observations?

- I** The toxin was stored in the central vacuole.
- II** The toxin cannot cross the membrane of the organelle in which it is stored.
- III** The toxin was stored in the chloroplast.
- IV** The toxin is likely to be lipid soluble.
- V** The toxin may be an enzyme.

- A** I, II and V
- B** I, IV and V
- C** II, III and IV
- D** III, IV and V

- 9** Which of the following curves best represent the movement of substances by simple diffusion and facilitated diffusion?



|          | simple diffusion | facilitated diffusion |
|----------|------------------|-----------------------|
| <b>A</b> | d                | b                     |
| <b>B</b> | d                | c                     |
| <b>C</b> | c                | a                     |
| <b>D</b> | c                | d                     |

- 10** The table describes some molecules found in a cell surface membrane.

Which of the following is correct?

|          | <b>cholesterol</b>                                    | <b>glycolipid</b>                 | <b>glycoprotein</b>                                 | <b>phospholipid</b>                     | <b>protein</b>                                        |
|----------|-------------------------------------------------------|-----------------------------------|-----------------------------------------------------|-----------------------------------------|-------------------------------------------------------|
| <b>A</b> | is a precursor of certain hormones                    | involved in active transport      | transports substances across the membrane           | is amphipathic in nature                | involved in DNA replication                           |
| <b>B</b> | regulates membrane fluidity at different temperatures | involved in cell-cell recognition | increases membrane fluidity at high temperature     | involved in DNA replication             | limits the permeability of the membrane               |
| <b>C</b> | decreases membrane fluidity at high temperature       | involved in cell-cell recognition | acts as a signal binding site during cell signaling | limits the permeability of the membrane | can change in conformation when phosphorylated by ATP |
| <b>D</b> | has a four ring structure                             | involved in active transport      | acts as a ligand binding site                       | is amphoteric in nature                 | limits the permeability of the membrane               |

- 11** The figures below show the complete karyotypes of 2 rodents of the same species. In this species of rodents, males are heterogametic.



Rodent A (Male)

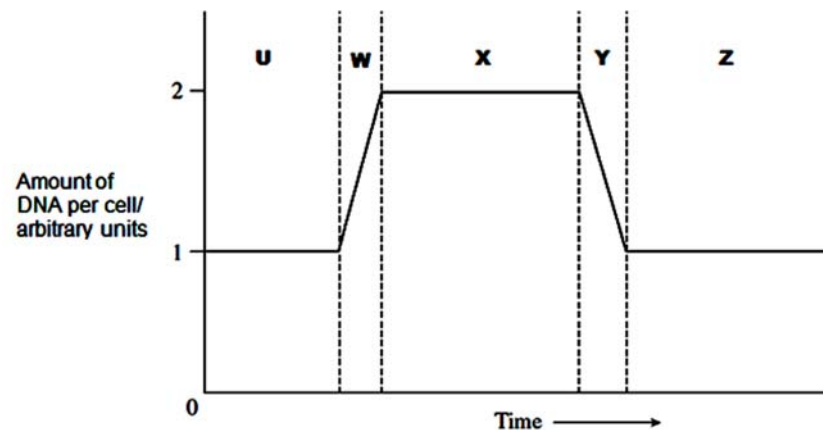


Rodent B (Female)

Which of the following observations is not true?

- A** A chromosomal aberration occurred in the ovary of the mother of rodent **B**.
- B** Rodent **A** is diploid and  $2n = 16$ .
- C** Rodent **B** has 1 missing X chromosome.
- D** Non-disjunction of autosomes occurred in rodent **B**.

- 12** The graph shows changes in the amount of DNA in a cell during one cell cycle. The letters U – Z marks out the different phases in the cell cycle.



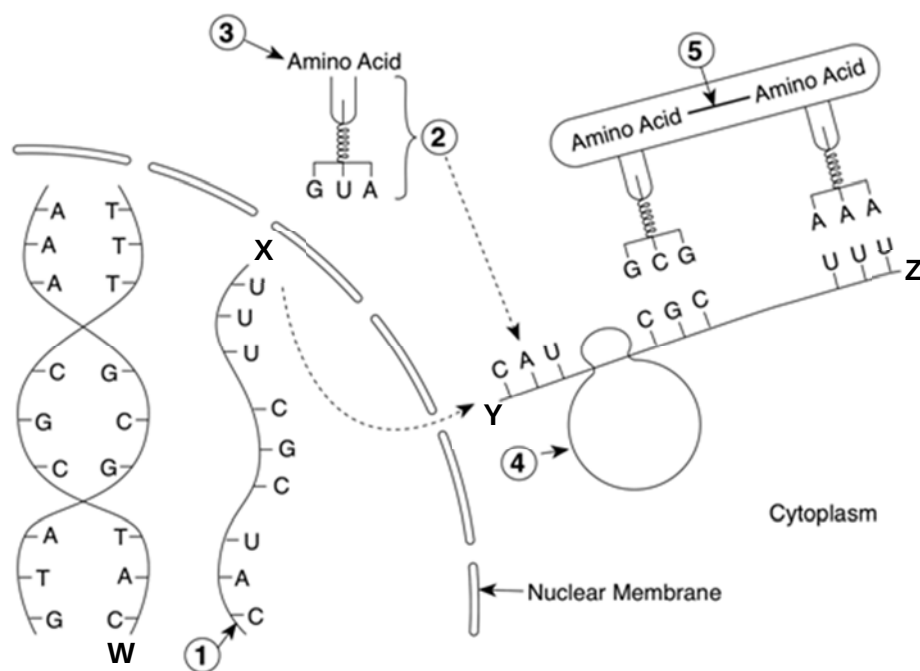
Many drugs that are used to treat cancer work at different time periods during the cell cycle.

- I One of the drugs, Cisplatin, binds to DNA and stops free DNA nucleotides from joining together.
- II Another drug, B, stops spindle fibres from shortening.

With reference to the cell cycle above, determine where these 2 drugs work.

|          | Cisplatin | Drug B |
|----------|-----------|--------|
| <b>A</b> | W         | X      |
| <b>B</b> | W         | Y      |
| <b>C</b> | U         | X      |
| <b>D</b> | U         | Z      |

13 The diagram below represents some biochemical reactions involved in protein synthesis.



Which is correct?

|   | Entire molecule coded directly from DNA is represented by | 5' end of molecule | Enzyme involved in catalysing bond 5 |
|---|-----------------------------------------------------------|--------------------|--------------------------------------|
| A | 1 and 2                                                   | Z                  | peptidyl transferase                 |
| B | 1 and 2                                                   | Y                  | aminoacyl tRNA synthetase            |
| C | 1, 2 and 3                                                | X                  | aminoacyl tRNA synthetase            |
| D | 1, 2 and 4                                                | W                  | peptidyl transferase                 |

- 14 A mutation of a gene coding for an ion pump in cell surface membranes results in the substitution of one amino acid, arginine, by another, histidine.

The DNA triplet codes for the two amino acids are shown.

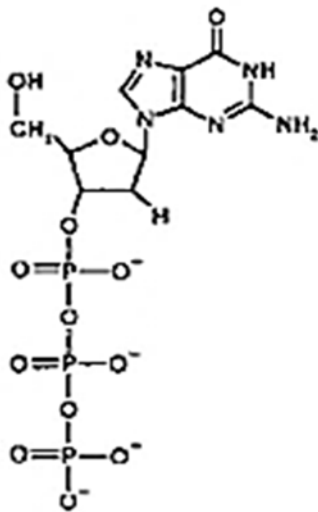
Arginine GCA  
GCG  
GCT  
GCC  
TCT  
TCC

Histidine GTA  
GTG

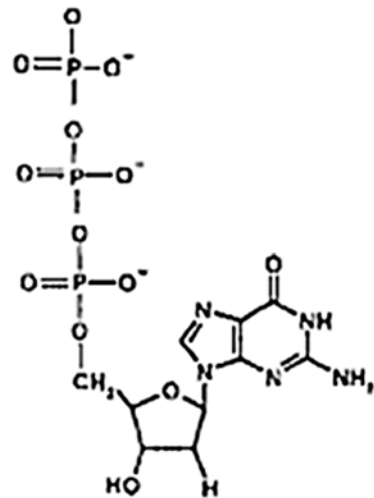
Which mutation has occurred in the DNA?

- A Addition of an extra nucleotide
- B Replacement of a nucleotide with a purine base by one with a different purine base
- C Replacement of a nucleotide with a purine base by one with a pyrimidine base
- D Replacement of a nucleotide with a pyrimidine base by one with a different pyrimidine base
- 15 Which of the following is the correct physiological substrate for DNA polymerase?

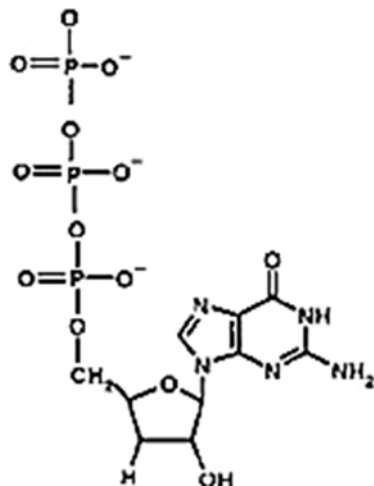
A



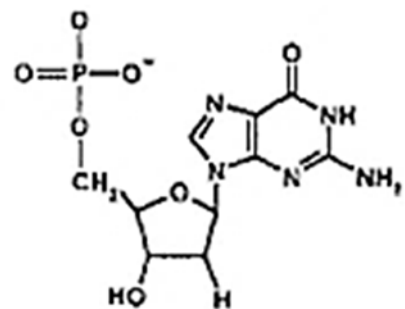
B



C



D



**16** The following events occur during transcription.

- 1 Bonds break between complementary bases.
- 2 Bonds form between complementary bases.
- 3 Sugar-phosphate bonds form.
- 4 Free nucleotides pair with complementary nucleotides.

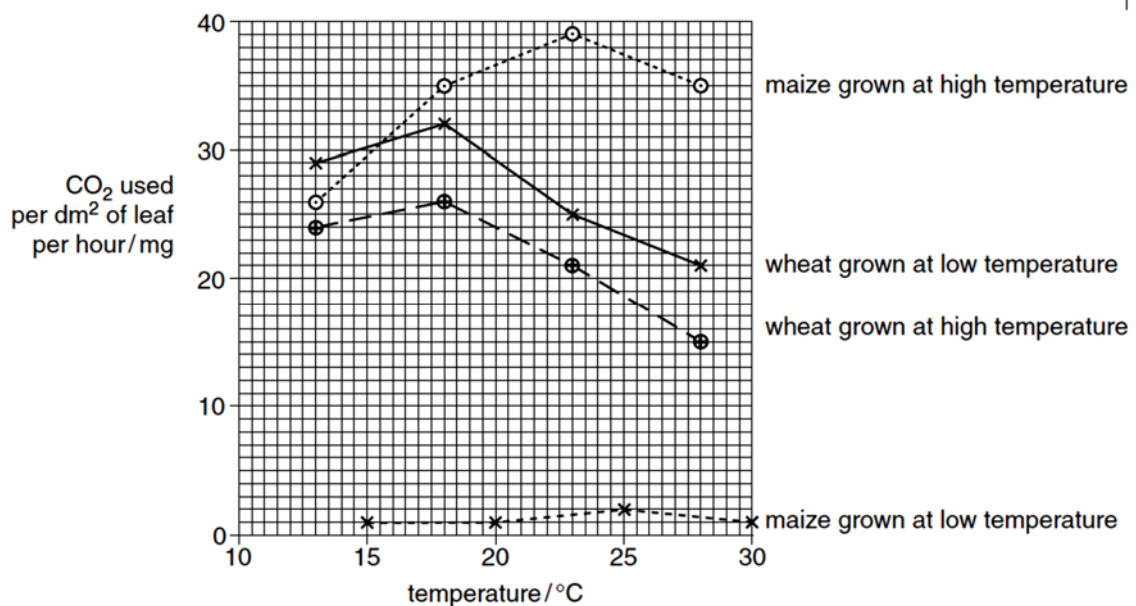
Before the mRNA leaves the nucleus, which events will have occurred twice?

- A** 1 and 2 only
- B** 1, 3 and 4 only
- C** 2, 3 and 4 only
- D** 1, 2, 3 and 4

**17** The coding region of a gene is 105 nucleotides long, including both the start and stop codons. Which would be the most likely effect of a single nucleotide deletion at position 76 in the coding region?

- A** Only the active site would be affected.
- B** The entire amino acid sequence of the polypeptide would change.
- C** There would be changes in only the first 25 amino acids.
- D** There would be changes in only the last 9 amino acids.

- 18 Young maize and wheat plants were grown to maturity at high and low temperatures. The rate of photosynthesis in each of these mature plants was measured at different temperatures. The rate of photosynthesis was measured as the amount of  $\text{CO}_2$  used per  $\text{dm}^2$  of leaf per hour. The results are shown in the graph below.



What information can be concluded from the graph above?

- For plants grown at high temperature, the rate of photosynthesis is optimum at 25°C in maize and 18°C in wheat.
  - For plants grown at high temperature, maize had a greater increase in rate of photosynthesis compared to wheat until optimum temperature was reached.
  - The rate of photosynthesis was affected more significantly in maize plants than in wheat plants grown at low temperatures.
  - Low temperatures slowed down the formation of membranes in maize plants but not in wheat plants.
- A 1 and 4 only
- B 2 only
- C 2 and 3 only
- D 3 and 4 only

- 19** The blue dye DCPIP can be converted to colourless DCPIP as shown below:



A suspension of chloroplasts was made by grinding fresh leaves in buffer solution and centrifuging the mixture. Tubes were then prepared and treated in the following ways.

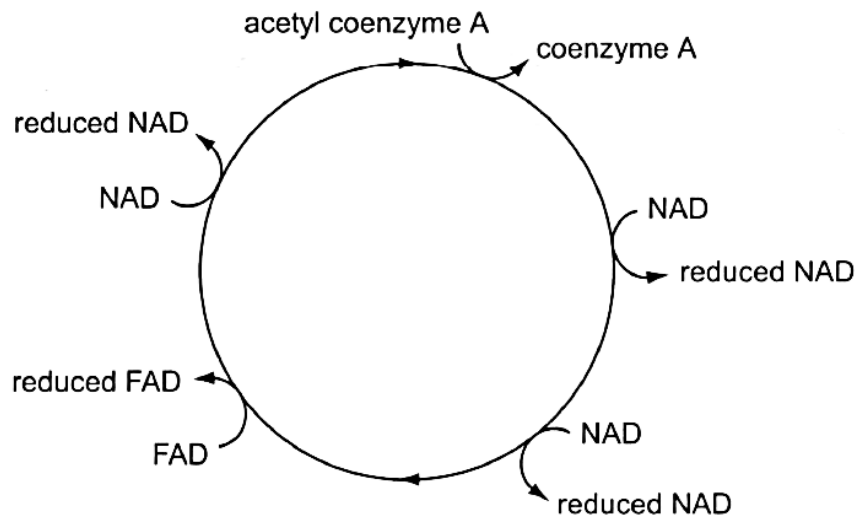
| tube     | contents                                                              | treatment            | colour     |                  |
|----------|-----------------------------------------------------------------------|----------------------|------------|------------------|
|          |                                                                       |                      | at start   | after 20 minutes |
| <b>1</b> | 1 cm <sup>3</sup> chloroplast suspension<br>+ 5 cm <sup>3</sup> DCPIP | illuminated strongly | blue green | green            |
| <b>2</b> | 1 cm <sup>3</sup> buffer solution<br>+ 5 cm <sup>3</sup> DCPIP        | illuminated strongly | blue       | blue             |
| <b>3</b> | 1 cm <sup>3</sup> chloroplast suspension<br>+ 5 cm <sup>3</sup> DCPIP | left in the dark     | blue green | blue green       |

Which statement is a possible conclusion for the observation above?

- A** Electron transfer from reduced NAD to DCPIP causes the decolourisation of DCPIP.
- B** NADP was oxidised and the electron was used to decolourise DCPIP.
- C** Light dependent reaction which occurs in the chloroplasts yield free electrons which reduced DCPIP.
- D** Either strong illumination or the buffer solution used in the extraction of chloroplasts could oxidise DCPIP.



- 20 The diagram shows the reactions of the hydrogen carriers in the Krebs cycle.

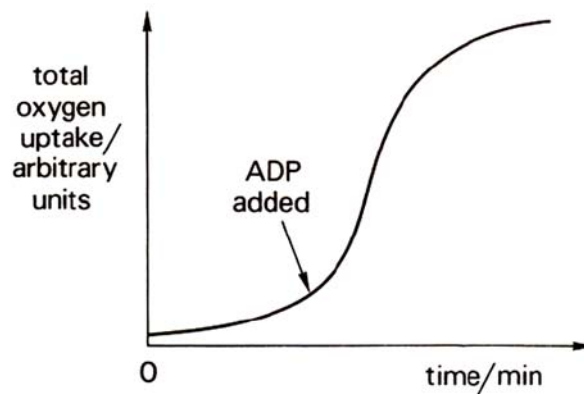


The average yield of ATP, in oxidative phosphorylation, is 3 molecules from each molecule of reduced NAD and 2 molecules from each molecule of reduced FAD.

What is the average yield of ATP, in oxidative phosphorylation, from the hydrogen carriers reduced in the Krebs cycle, from one molecule of glucose?

- A 11
- B 22
- C 34
- D 36

- 21 The graph shows the total oxygen uptake by a sample of homogenised liver tissue over a period of time.



Which statement explains the sharp rise in oxygen uptake after the addition of ADP?

- A Electron transport is coupled to the phosphorylation of ADP.
  - B More protons move directly across the phospholipid bilayer of the cristae, generating the proton pool in the intermembrane space.
  - C Slower utilisation of reduced NAD and FAD causes molecular oxygen to be reduced to water.
  - D Oxidative phosphorylation occurs as protons are pumped back into the matrix through ATP synthase, phosphorylating ADP.
- 22 A plant is known to be heterozygous at two gene loci. The pollen grains from this plant are used to fertilise another plant of the same genotype.

What is the probability that an embryo will be homozygous dominant at one locus?

- A 1 in 4
- B 3 in 8
- C 5 in 8
- D 1 in 16

- 23** Fruit flies (*Drosophila*), homozygous for long wings, were crossed with fruit flies homozygous for vestigial wings. The F<sub>1</sub> and F<sub>2</sub> generations were raised at three different temperatures.

At each temperature, the F<sub>1</sub> generation all had long wings.

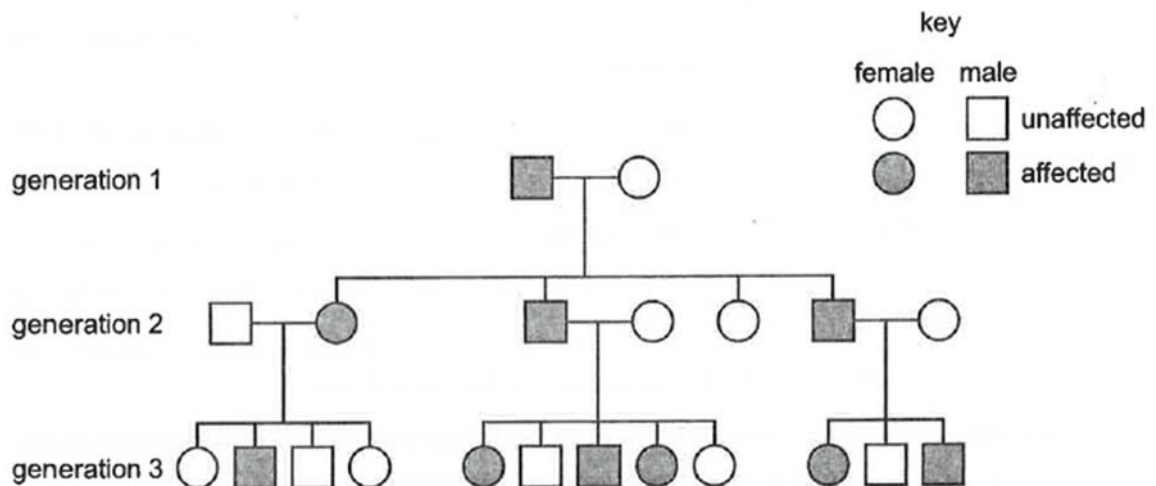
The table shows the results in the F<sub>2</sub> generation.

| temperature / °C | result                                                           |
|------------------|------------------------------------------------------------------|
| 21               | $\frac{3}{4}$ long wings, $\frac{1}{4}$ vestigial wings          |
| 26               | $\frac{3}{4}$ long wings, $\frac{1}{4}$ intermediate wing length |
| 31               | all long wings                                                   |

Which statement explains these results?

- A** Heterozygous flies have vestigial wings only at 21°C or below but have long wings at 31°C or above.
- B** Long wing and vestigial wing illustrate codominance at 26°C.
- C** Long wing is dominant at higher temperatures but vestigial wing is dominant at lower temperatures.
- D** Vestigial wing is recessive but causes a vestigial wing phenotype only at lower temperatures.

- 24** The pedigree shows the inheritance of a genetic condition in a family for three generations.



Which evidence indicates that this genetic condition is autosomal?

- A** Affected females always have affected sons.
- B** Affected males do not pass it on to their sons.
- C** Affected parents always have affected offspring.
- D** Males and females are equally affected.

- 25** A strain of mice have been genetically modified so that they display symptoms of a recessive sex-linked trait that causes death before the mice are able to breed.

In these mice, 33% of the alleles of this gene were recessive.

In multiple crossings of carrier females and normal males, it was found that 50% of the males died but no females.

What percentage of the alleles of this gene in the surviving offspring would be recessive?

- A** 0 %
- B** 17 %
- C** 20 %
- D** 33 %

- 26** Healthy, embryonic stem cells offer great promise to the cure of many diseases because \_\_\_\_\_.

- I** They are totipotent.
- II** They are more readily available than other cells.
- III** They can be easily cultured *in vitro*.
- IV** They will differentiate in the presence of appropriate molecular signals and produce cells which are normal.

Which of the above statement(s) is/are true?

- A** II, III and IV
- B** I, III and IV
- C** I and IV
- D** IV only

- 27** Animals with horizontal stripes are bitten less frequently by tsetse flies. The flies carry diseases that infect zebras.

Which explains how zebras might evolve to have more horizontal stripes?

- A** Bites from tsetse flies cause mutations. If a zebra has a mutation it will die and not pass its genes onto its offspring which will not have more horizontal stripes.
- B** If two zebras with horizontal stripes mate, their offspring will have more horizontal stripes. Horizontal stripes will become dominant. This is natural selection.
- C** Tsetse flies are a selection pressure. The zebras would gradually develop more horizontal stripes and pass them on to their offspring so that are not bitten by flies.
- D** Zebras with more horizontal stripes get fewer diseases from tsetse flies. These zebras live longer and breed more, passing the allele for more horizontal stripes to their offspring.

- 28** Bacteria in the genus *Wolbachia* infect many butterfly species. They are passed from one generation to the next in eggs, but not in sperm, and they selectively kill developing male embryos.

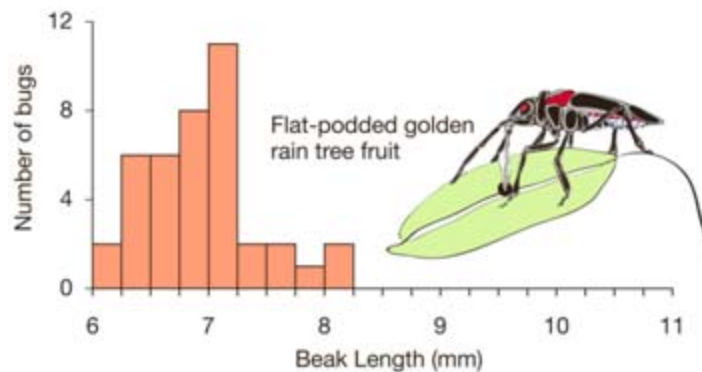
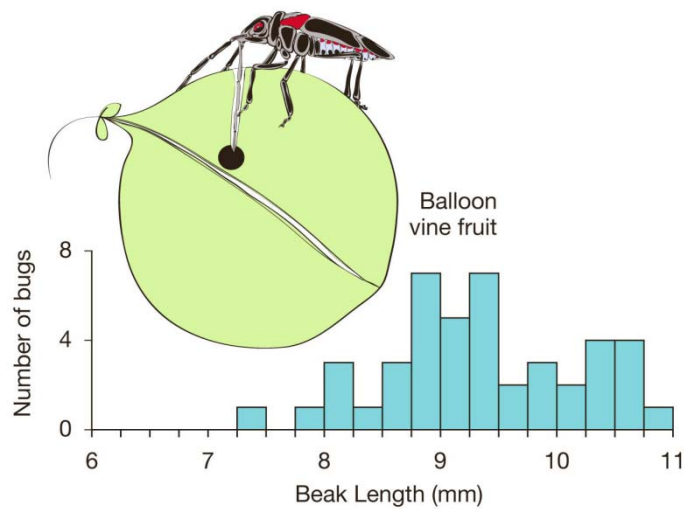
In Samoa in the 1960s, the proportion of male blue moon butterflies fell to less than 1% of the population. However, by 2006, the proportion of males was almost 50% of the population.

Resistance to *Wolbachia* is the result of the dominant allele of a suppressor gene.

Which statements correctly describe the evolution of resistance to *Wolbachia* in the blue moon butterfly population?

- I** *Wolbachia* acts as a selective agent.
  - II** The selective killing of male embryos is an example of artificial selection.
  - III** When infected with *Wolbachia*, male embryos that are homozygous for the recessive allele of the suppressor gene die.
  - IV** All male embryos that carry the dominant allele of the suppressor gene pass that allele to their offspring.
  - V** The frequency of the dominant allele of the suppressor gene rises in the butterfly population.
- A** I and IV
  - B** I, III and V
  - C** II and III
  - D** II, IV and V

- 29** Soapberry bugs in Florida originally fed on the native balloon vine, using their sharp beaks to penetrate the fruits to reach the seeds. In the 1920s, the flat-podded golden rain tree was introduced from Asia to mainland central Florida. This has thinner-skinned fruit, and correspondingly, soapberry bugs evolved shorter beaks after they switched completely to feed on this new host plant.



What best describes the type of evolutionary process depicted above?

The soapberry bugs are undergoing

- A** stabilizing selection.
- B** genetic drift.
- C** natural selection.
- D** directional selection.

- 30 Which of the following about global warming is correct?
- A growing of crops → burning of trees to clear land → increased methane production → global warming
  - B global warming → melting of permafrost → emergence of deadly diseases → possibility of increased mortality
  - C global warming → metabolism of insects in the tropics increases → size of insects increases → amount of pesticides required to kill insect pests increases
  - D global warming → calcium carbonate uptake by corals increases → coral bleaching → disruption of food web in ocean

**- END OF PAPER -**

|          |   |           |   |           |   |           |   |           |   |           |   |
|----------|---|-----------|---|-----------|---|-----------|---|-----------|---|-----------|---|
| <b>1</b> | B | <b>6</b>  | C | <b>11</b> | D | <b>16</b> | A | <b>21</b> | A | <b>26</b> | D |
| <b>2</b> | B | <b>7</b>  | C | <b>12</b> | A | <b>17</b> | D | <b>22</b> | B | <b>27</b> | D |
| <b>3</b> | C | <b>8</b>  | A | <b>13</b> | A | <b>18</b> | B | <b>23</b> | D | <b>28</b> | B |
| <b>4</b> | C | <b>9</b>  | B | <b>14</b> | D | <b>19</b> | C | <b>24</b> | D | <b>29</b> | D |
| <b>5</b> | B | <b>10</b> | C | <b>15</b> | B | <b>20</b> | B | <b>25</b> | C | <b>30</b> | B |